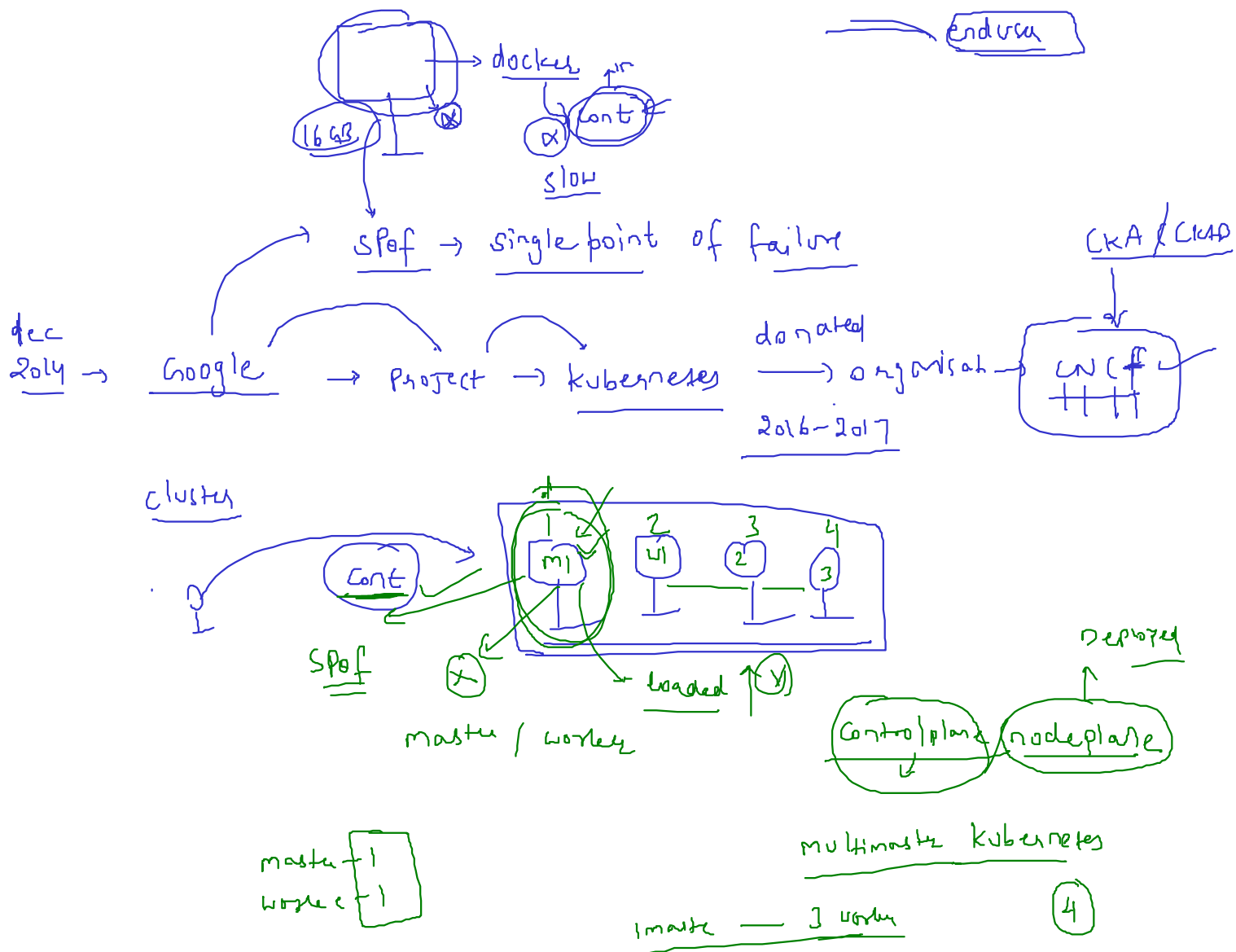


Recap :-

1. Phases of web deployment
2. What is containerization technology
3. How it is working
4. Whole architecture
5. Basic command of docker
 - in base machine
 - in containers
6. apache configure
7. port mapping
8. Dockerfile
 - a. COPY , FROM , label , RUN , EXPOSE
 - CMD , WORKDIR

1. Disadvantages of docker
2. Why kubernetes came
3. architecture of kubernetes
4. components
5. Create the cluster
6. Access the cluster
7. Kubernetes Resources
 - a. Namespace
 - b. Pods
 - c. Deployments
 - d. Services

- i. ClusterIP
 - ii. NodePort
 - iii. LoadBalancer
 - iv. External Name
- x

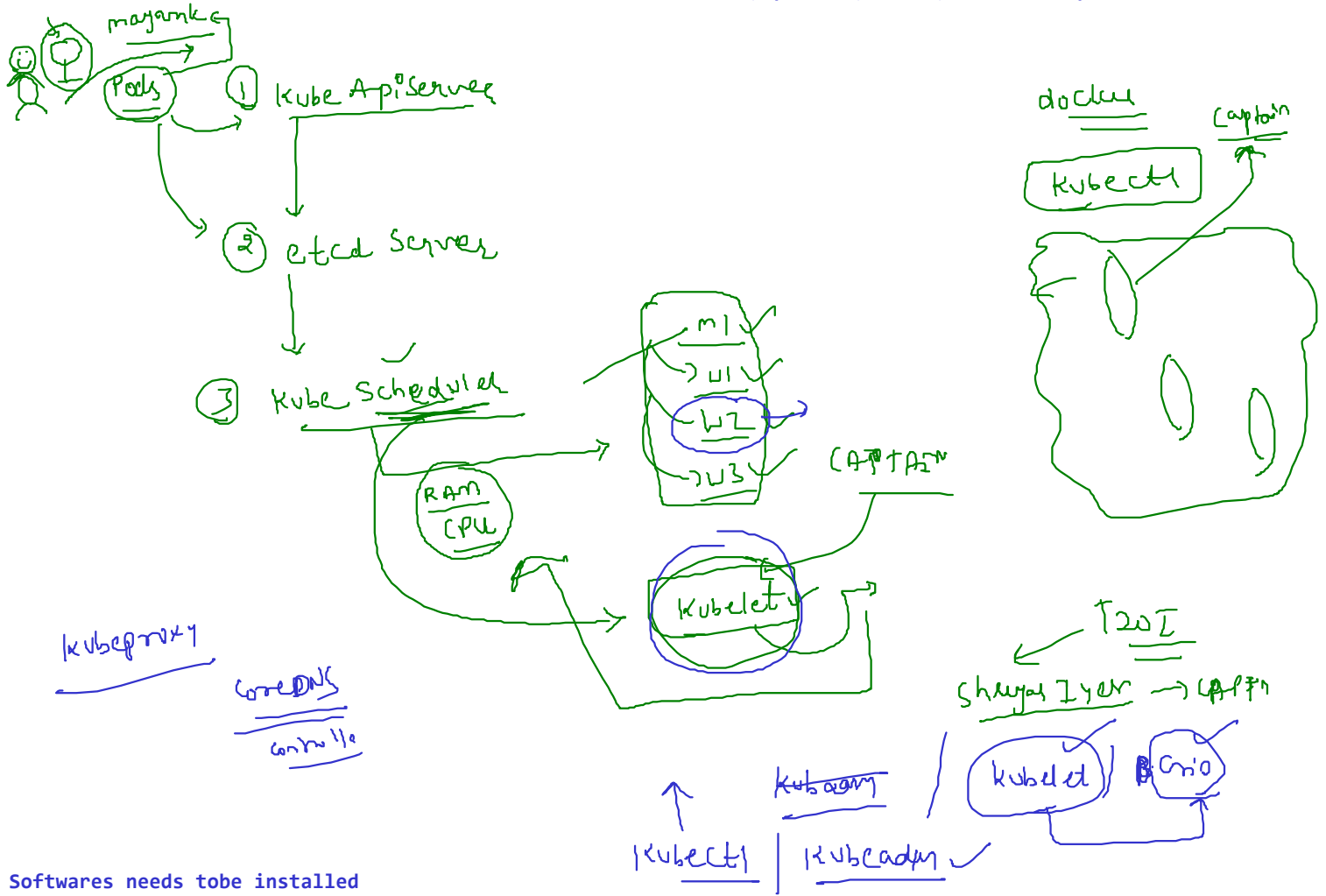


Components of kubernetes

Docker is a container creation tool

Kubernetes is not a container creation tool its a orchestration tool / management tool of containers

docker, podman , crio , buildahs pods -- >>> containers



Softwares needs to be installed

1. `crio` ,
2. `kubelet`
3. `kubect1`
4. `kubeadm`

Components of kubernetes

kube api server . etcd server , kube scheduler , coredns kubeproxy

~~US K~~

Pre req.

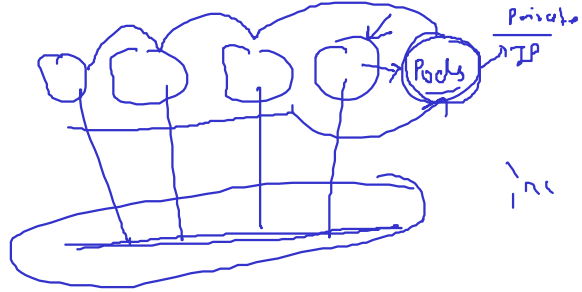
1. Swap memory - does not support - means turn off
2. kernel module - br_netfilter and overlay2
load
enable
3. Enable IPv4 forwarding
4. install 4 softwares
a. crio , b. kubelet , c.kubectrl , d. kubeadm

kubeadm init

Certificate Based Login

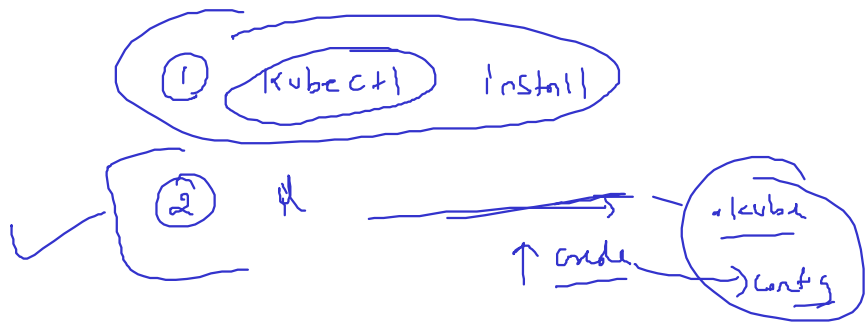
→ file

④



N/W solution

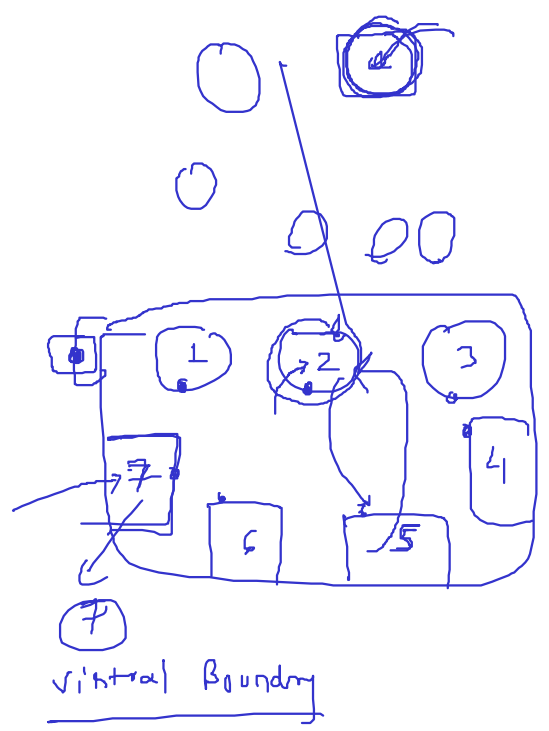
Calico
Flannel
Weave-net



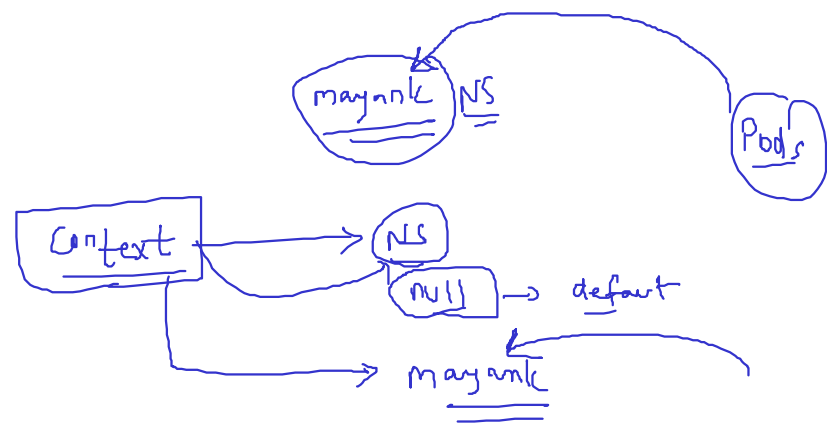
Smallest unit/resource/kind = Pods

namespace

1 key → No. 2
2 keys → main No. 2

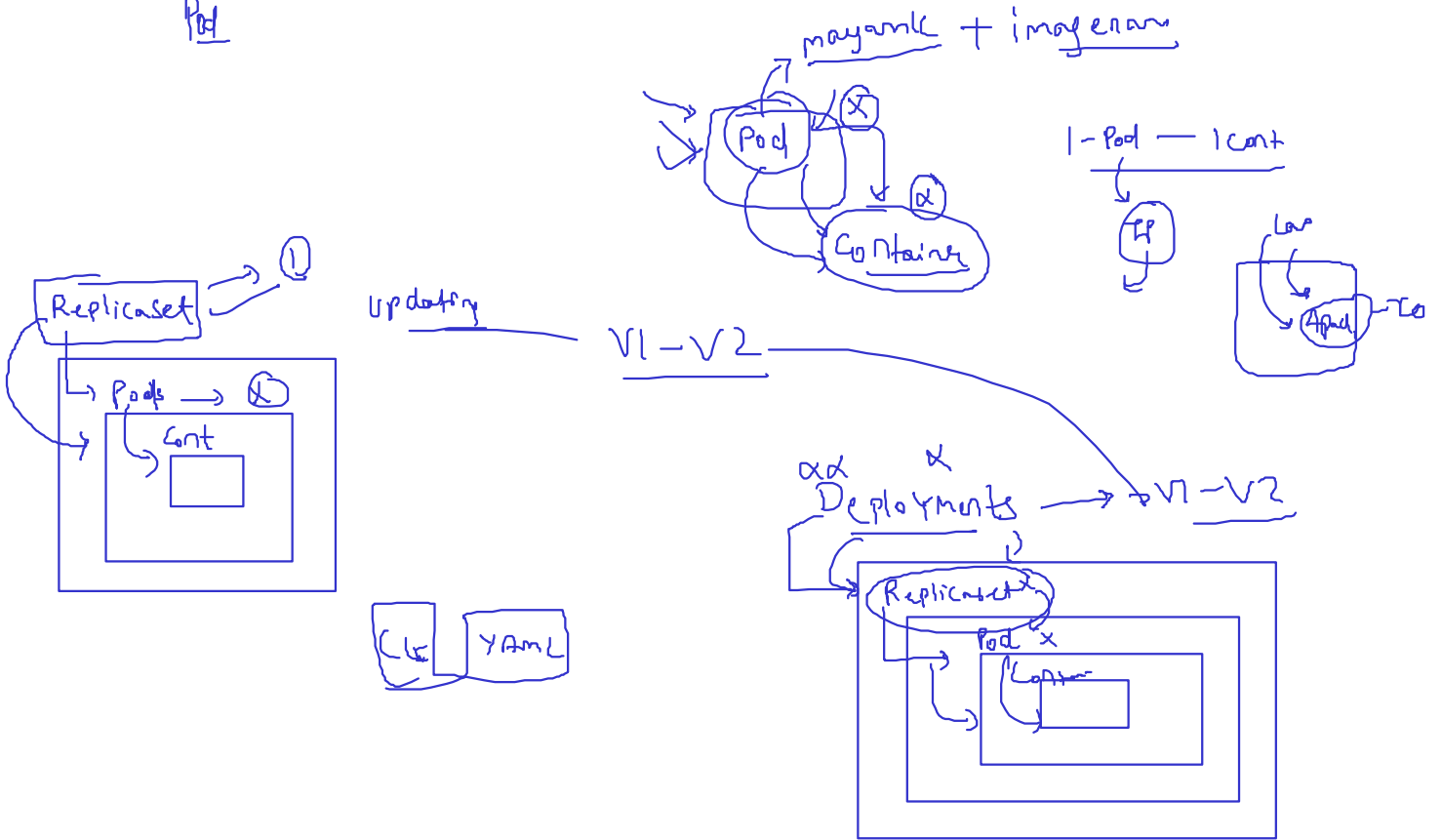


4





Pod

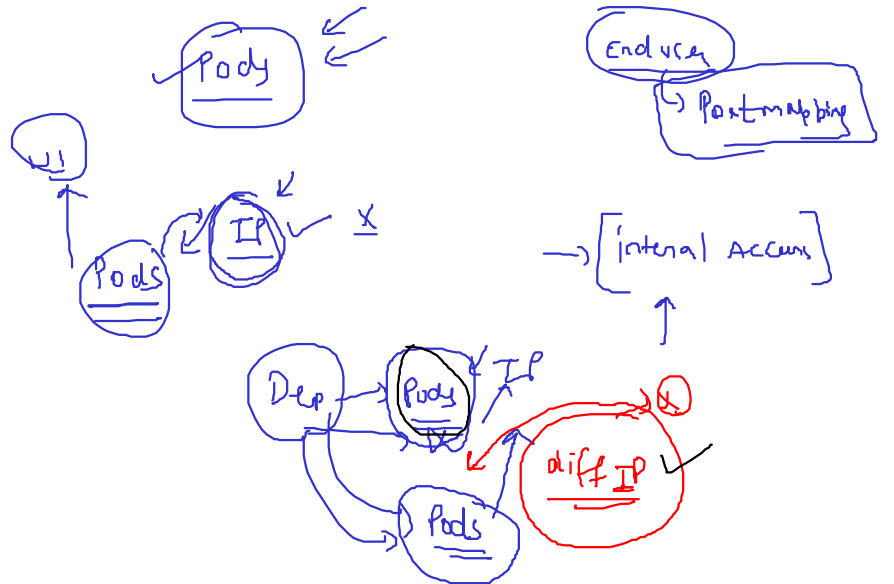


YAML → Pod → rolling → 2-3 mins

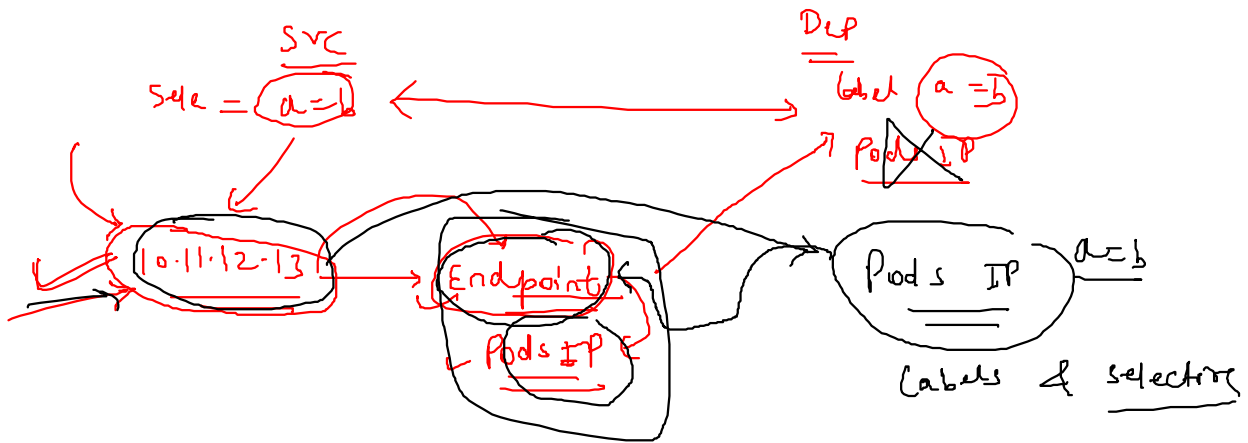
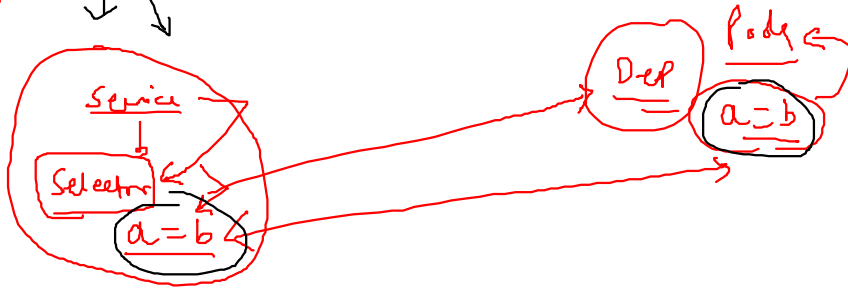
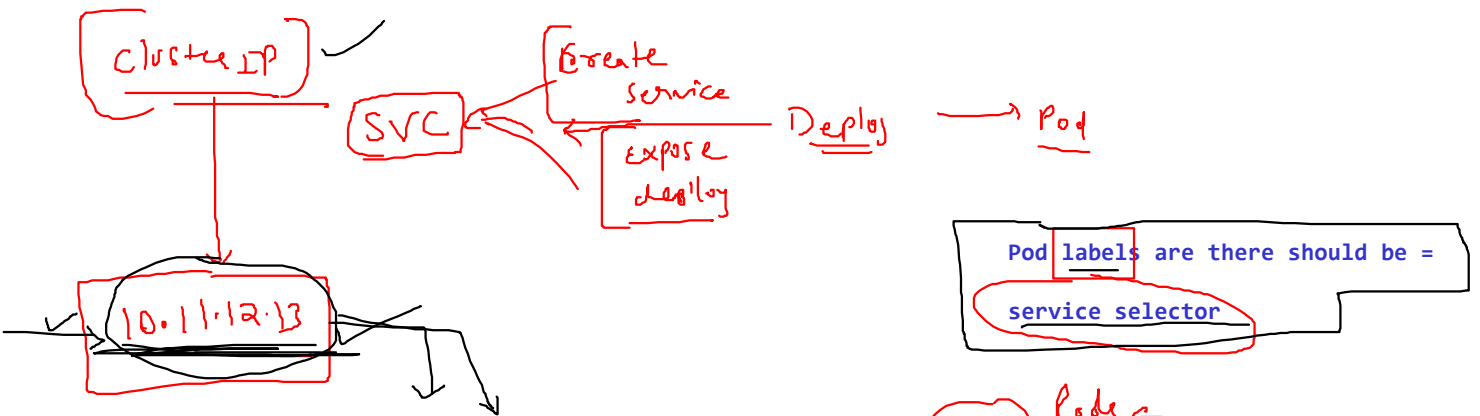
deployment YAML → rolling → 3 mins

- ① write it 3-5
- ② git diff / search LP Path → 1-2 mins

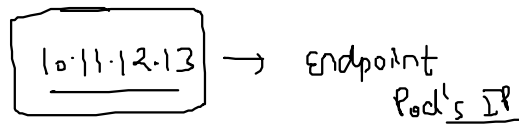
③ Generate YAML → lose Conds



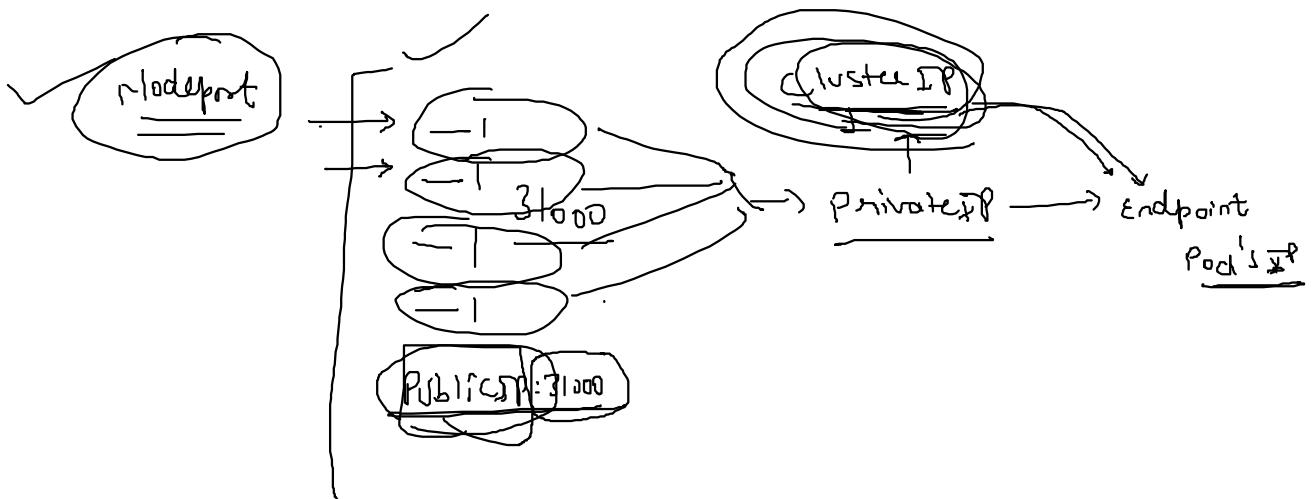
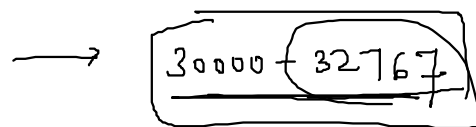
Internal Access Reliable Source

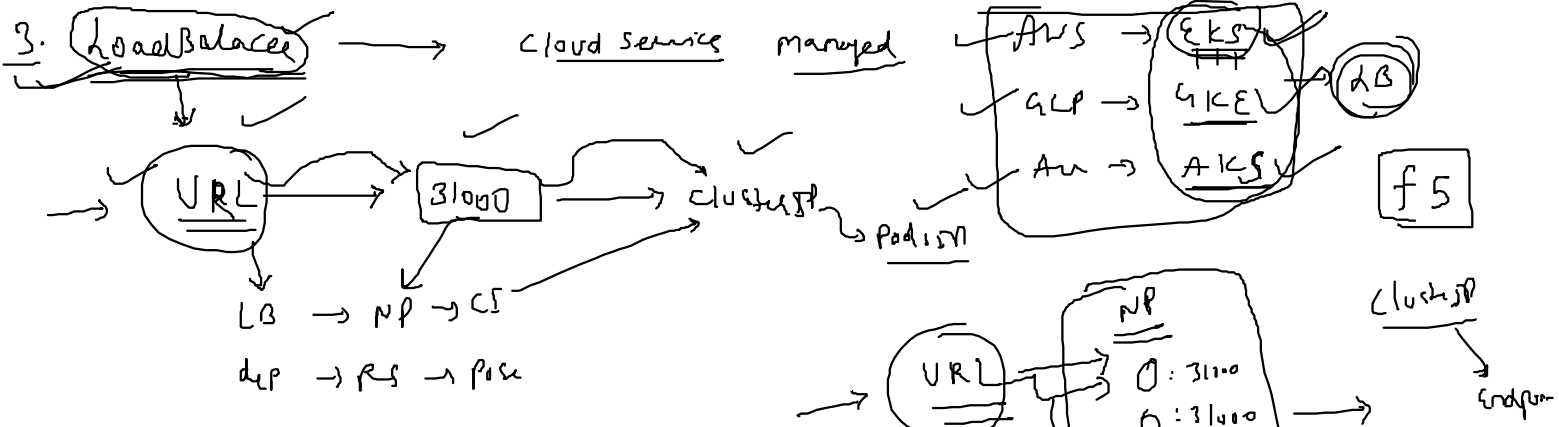


Cluster IP



2. Nodeport ^{Port No.}





4. external name

